

ADR-0004 — PostgreSQL for metadata; audio files behind a storage abstraction

ACCEPTED Date: 2026-09-01 · Deciders: Nicolà Widmer

Context

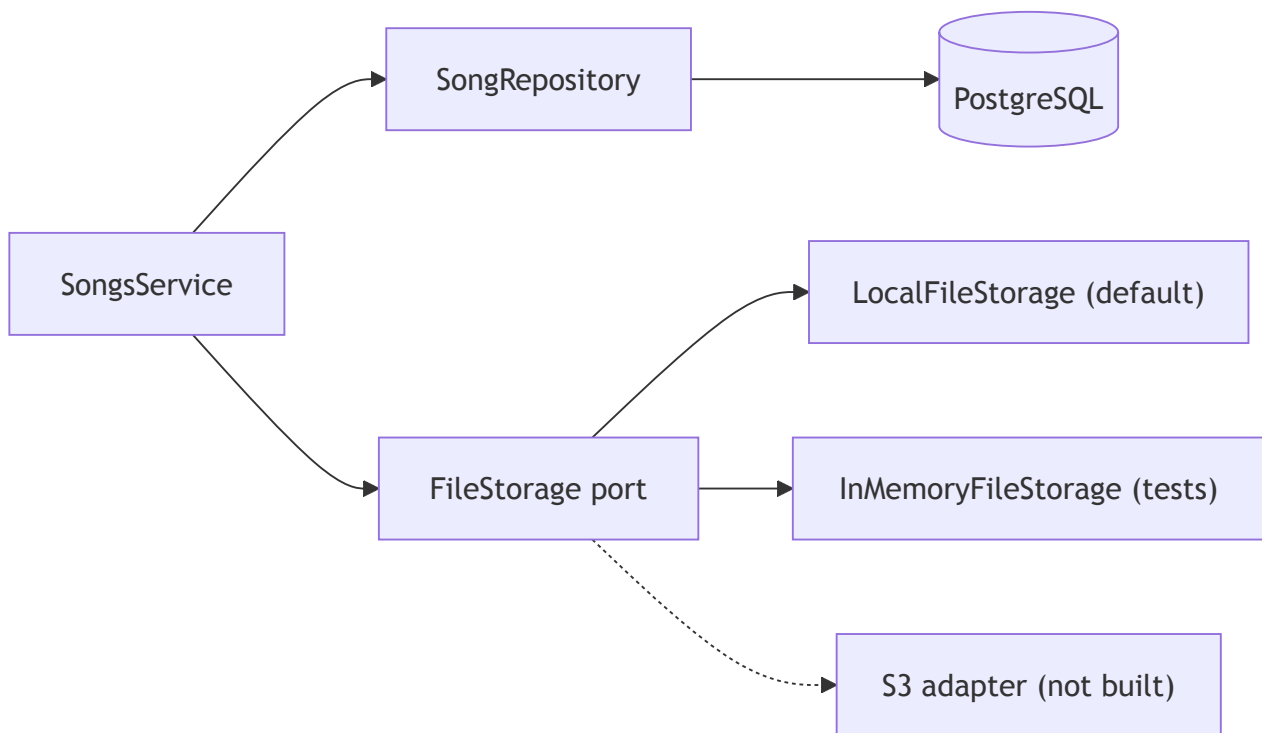
Two kinds of data: small records (songs, playlists, users) and MP3s up to 20 MB. The default must run without a cloud account. S3 is an optional extra. Audio needs range reads.

Options

1. **PostgreSQL for metadata, audio behind a FileStorage port**
2. Store MP3s in PostgreSQL (BYTEA)
3. Call the filesystem directly, add S3 later

Decision

Option 1.



A song row stores an opaque key. Delete order: row first, then files. A failed file delete leaves an orphan file, which is harmless.

Consequences

- Good: use cases don't know where bytes live.
- Good: the database and its backups stay small.
- Bad: two stores can drift (orphan files).
- Bad: each adapter needs its own tests.
- Rejected BYTEA: bloats the database and its backups.
- Rejected direct filesystem: S3 would later leak into use cases.